Advanced Kerr-Microscopy-Based MOKE Magnetometry for the Anisotropy Characterisation of Magnetic Films

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Abstract overview

- Advanced wide-field Kerr microscopy and magnetometry approach for magnetic anisotropy investigation in thin films.
- Analytical and experimental analysis of longitudinal Kerr contrast versus light polarisation and incidence plane.
- Fully automated procedure for measuring magnetisation reversal loops.
- No mechanical adjustment of the setup required.

Why magnetic anisotropy matters

- Magnetisation reversal loops as a function of external field are key for fundamental and applied magnetism.
- Dependence of reversal on magnetic field angle directly reflects material anisotropy.
- Measurements require the magnetometer sensitivity to be aligned with the field direction for all angles.

Limitations of conventional methods

- Vibrating sample magnetometry (VSM) needs mechanical rotation of the sample.
- Anisotropic magnetoresistance techniques require lithography and patterning of the sample.
- Mechanical adjustments introduce experimental errors and spurious signals.

Magneto-optical Kerr effect (MOKE)

- MOKE: rotation of the polarisation plane of linearly polarised light upon reflection from a magnetic, reflective sample.
- Three basic Kerr geometries:
  - Polar: magnetisation out of plane.
  - Longitudinal: in-plane, along incidence plane.
  - Transversal: in-plane, transverse to incidence plane.

Wide-field Kerr microscopy

- Effective tool for magnetic domain imaging and magnetometry.
- Kerr intensity from a freely selectable image area serves as the magnetometer signal.
- Conventional wide-field systems typically require sample rotation for angular dependence.

Novel aspects of the approach

- Sensitivity direction for p-polarised light can deviate from the incidence plane.
- This deviation is usually neglected in standard MOKE systems.
- Magnetic anisotropy in modern sensors can be influenced and tuned.
- In magnetostrictive materials, anisotropy can be controlled by mechanical stress.

Experimental setup overview

- Wide-field Kerr microscope.
- Quadrupole electromagnet for fields in arbitrary in-plane directions.
- Sample holder for controlled mechanical stress application.

Microscope optical system

- Wide-field Kerr microscope based on Carl Zeiss Axio Scope.
- Light from eight LEDs guided independently via glass fibres.
- Fibre ends arranged in a cross-like geometry in the conoscopic plane.
- Incidence plane defined by selecting appropriate LEDs.

Quadrupole magnet system

- Quadrupole electromagnet generates in-plane magnetic fields in arbitrary directions.
- Two orthogonal pairs of solenoids produce the field components.
- Field direction can be changed without rotating the sample.

Stress application holder

- Sample holder with asymmetrically cut copper block and wedge.
- Displacement of the lower block along a guiding screw moves the wedge upward.
- Substrate bending introduces tensile stress in the magnetic film at the surface.
- The applied tensile stress direction is well-defined.

Wide-field microscope light source

- Special light source with eight independently driven LEDs.
- LED light guided to the microscope by glass fibres.
- Cross arrangement of fibre ends defines possible incidence directions.

LED control and domain contrast

- Incidence direction set by software-controlled LED selection.
- Sequential pulsing of opposite LEDs and image subtraction yields pure in-plane domain contrast.
- Faraday contributions in the microscope lenses are effectively cancelled.

Vectorial sensitivity concept

- Orthogonal LED pairs are driven in pulsed mode synchronised with the camera.
- Enables real-time two-component imaging of magnetisation.
- Longitudinal Kerr measurements at two orthogonal incidence planes.
- Provides simultaneous access to magnetisation components along different directions.

Light–magnetisation interaction

- Reflected linearly polarised light rotates clockwise or counter-clockwise depending on magnetisation orientation.
- Regular reflected amplitude A_N and Kerr amplitude A_K are mutually perpendicular.
- Interference of A_N and A_K gives total amplitude $A = A_N + A_K$.

Analyser configuration and contrast

- Domain contrast produced by blocking reflected light from one magnetisation state.
- Analyser is set almost perpendicular to the polariser with a small opening angle.
- Motorised analyser allows precise electronic control; polariser is manually adjusted.

Sensitivity direction analysis

- Sensitivity direction: in-plane direction with maximum MOKE contrast.
- For s-polarised light, sensitivity always lies within the incidence plane.
- For p-polarised light, the direction of maximum contrast can deviate from the incidence plane.
- This effect is crucial for quantitative magnetometry.

Intensity vs field angle

- For s-polarised light along the y-axis, integral image intensity follows a sinusoidal dependence with phase shift $\Delta\theta = 0^\circ$.
- For p-polarised light along the x-axis, intensity can be approximated by a cosine with $\Delta\theta \approx -16^\circ$.
- Demonstrates strong polarisation dependence of the angular response.

Deviation of sensitivity direction

- S-polarised configurations show no deviation of sensitivity from the incidence plane.
- For p-polarised light, the sensitivity direction strongly depends on analyser opening angle.
- Deviations can reach several tens of degrees for typical analyser settings.

Experimental Fe/FeO sample

- Nominally 10 nm thick Fe/FeO film.
- Prepared by DC magnetron sputtering on $Au/Cr/SiO_2/Si(100)$ substrate.
- External field of 15 mT used to saturate the film and align magnetisation with the field.
- Measurements verify theoretical predictions on sensitivity direction deviations.

Anisotropy measurement strategy

- Angle-dependent image intensity recorded with complementary sensitivities during field rotation.
- Magnetisation reversal loops acquired for different field angles relative to the sample.
- Conventional approach: field curves along many sample directions via stepwise sample rotation.

Automated anisotropy measurement

- Knowledge of exact sensitivity orientations enables reconstruction of loops along arbitrary in-plane field directions.
- Only two loops with complementary sensitivities are required.
- Provides a fast and efficient magnetometry scheme for anisotropy characterisation.
- No mechanical adjustment or sample displacement is needed.

Vector magnetisation reconstruction

- Reconstruction uses magnetisation projections on two sensitivity directions, e.g. S_x and S_y.
- Grey level of the image encodes average magnetisation projection along the respective sensitivity.
- From the two projections, magnitude and direction of magnetisation along the field direction are obtained.

Example hysteresis loops

- Hysteresis loop measured with s-polarised light along the y-axis.
- Corresponding loop measured with p-polarised light along the x-axis.
- Both loops recorded simultaneously within a single field scan in pure in-plane mode.
- Measurement time reduced from hours to minutes.

Determining sensitivity directions

- Absolute grey-level difference ($I^+ - I^-$) evaluated for orthogonal light incidences.
- Data points from loops with fields swept along fixed directions collected for angles between 0° and 180° with 2° steps.
- Additional data from continuous field rotation covering 0° to 360°.
- Combined data provide complete sensitivity-direction information.

Reconstructed magnetisation loops

- Magnetisation reversal loops reconstructed for selected field directions using the vector method.
- Coercive field H_c extracted and marked for each direction.
- Remanent magnetisation M_r/M_s obtained for all angles.
- Enables mapping of magnetic properties at any field orientation.

Angular dependence of anisotropy

- Angular dependence of coercive field H_c shows two broad peaks.
- Remanence diagram M_r/M_s exhibits a characteristic angular shape.
- Results indicate uniaxial in-plane magnetic anisotropy of the iron film.
- Easy axis at $\varphi \approx 110^\circ$, hard axis at $\varphi \approx 20^\circ$ in the sample coordinates.

Domain formation and blocking

- Large coercive fields around the hard-axis direction arise from blocked domain states.
- Blocked states consist of Néel-type domain walls stabilised by dipolar interaction.
- Understanding these mechanisms is essential for tailoring magnetic materials and devices.

Stress manipulation of anisotropy

- 100 nm NiFe (Permalloy) film used to study stress-induced anisotropy tuning.
- Remanence M_r/M_s versus angle without applied stress shows clear uniaxial anisotropy at about 70° to the x-axis.
- After applying tensile stress along the x-axis, the uniaxial anisotropy axis rotates towards the stress direction.
- Confirms a positive magnetostriction constant for the NiFe film.

Mechanism of stress-induced tuning

- Substrate bending forms a three-point bending configuration.
- Region of interest positioned above the movable copper wedge tip.
- Upward motion of the wedge forces the substrate to bow upward.
- Produces tensile stress along the x-axis in the magnetic film; by the Villari effect, anisotropy aligns with the stress.

Advantages over VSM and others

- VSM is an integral technique; MOKE magnetometry probes local areas independently.
- Wide-field Kerr microscopy allows free choice of area size from centimetres down to micrometres.
- Complex sample manipulation equipment can be added without affecting the optical scheme.
- No lithography or patterning of the sample is required.

Key findings

- Sensitivity direction for p-polarised light can deviate significantly from the incidence plane and depends on analyser setting.
- Fully automated procedure for magnetisation loop measurements along arbitrary in-plane field directions.
- Transverse-to-longitudinal Kerr coefficient ratio K_t/K_l is approximately unity for the investigated materials.

Conclusion

- Presented an advanced wide-field Kerr microscopy and magnetometry approach for thin-film anisotropy studies.
- Method is fully automated and does not require mechanical adjustment or sample displacement.
- Demonstrated functionality on Fe/FeO and NiFe thin magnetic films.
- Stress-induced tuning of anisotropy opens new routes for optimising magnetic sensors.

Applications and outlook

- Direct applications in fundamental research on magnetic thin films.
- Particularly suited for materials with uniaxial in-plane anisotropy.
- Stress-tunable anisotropy relevant for advanced magnetic sensors and devices.
- Future work: extension to multilayers and magnetically frustrated systems.

Thank you!

Questions are welcome.